

Bank of Leeds Strategic Infrastructure Bond Proposal

Humber Tidal Energy & Flood Defence Infrastructure Bond

A 30-Year Regional Infrastructure Bond to Finance Yorkshire's Largest Renewable Energy Asset

Prepared for:

Bank of Leeds – Infrastructure Finance Division

Investment Classification:

Long-Term Infrastructure Bond / Renewable Energy Asset / Regional Development

Proposed Instrument:

Bank of Leeds Humber Tidal Infrastructure Bond (30-Year)

Indicative Bond Size:

£2 billion initial issuance

Total Project Cost:

£10–12 billion

Investment Horizon:

30 years (bond maturity)

120+ years (asset life)

Executive Summary

The Bank of Leeds proposes establishing a dedicated financing vehicle to explore and potentially fund the development of a **Humber Tidal Barrage and Renewable Energy Gateway**.

The project would represent one of the largest infrastructure investments ever undertaken in Yorkshire, combining:

- tidal renewable electricity generation,
- flood defence,
- industrial decarbonisation,

- hydrogen production,
- transport infrastructure,
- regional economic regeneration.

The proposed financing structure would utilise a:

30-Year Bank of Leeds Infrastructure Bond

to attract:

- pension funds,
- insurance companies,
- institutional investors,
- sovereign infrastructure investors.

The underlying investment thesis:

"The Humber is not simply a waterway; it is a strategic energy and industrial asset capable of supporting Yorkshire's next century of economic growth."

The UK Government recognises tidal range projects as a potential component of future energy infrastructure, although proposals must demonstrate value, financial viability and environmental management.

1. Investment Opportunity

The Humber Energy Gateway

The Humber estuary is one of Britain's most important industrial regions.

It contains:

- Hull,
- Goole,
- Immingham,
- Grimsby,
- major chemical industries,
- ports,
- manufacturing,
- offshore wind supply chains.

A tidal barrage would convert this geographic asset into a:

Renewable Energy Infrastructure Zone

Strategic Objectives

The project would deliver:

Energy Security

Predictable renewable electricity generation.

Unlike wind and solar, tidal generation follows astronomical cycles and is highly predictable.

Industrial Competitiveness

Provide:

- lower-cost renewable electricity,
 - hydrogen production,
 - industrial decarbonisation.
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Climate Adaptation

Provide:

- tidal surge protection,
 - coastal resilience,
 - flood management.
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Regional Growth

Create:

- construction employment,
 - engineering capability,
 - manufacturing supply chains.
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2. Proposed Project Structure

Humber Tidal Infrastructure Corporation

A special-purpose company established by:

- Bank of Leeds,
- UK Government,
- infrastructure investors,

- energy companies.

Assets owned:

1. Humber tidal barrage
2. Turbine infrastructure
3. Grid connections
4. Hydrogen facilities
5. Associated transport infrastructure

3. Engineering Concept

Humber Tidal Barrage

Potential configurations:

Option A — Inner Humber Barrage

Purpose:

- energy,
- flood protection,
- urban resilience.

Option B — Outer Humber Energy Barrier

Purpose:

- larger electricity generation,
- transport crossing,
- industrial infrastructure.

Indicative Specification

Feature	Estimate
Barrage length	15–30 km
Installed capacity	1.0–1.5 GW
Turbines	100–200
Operational life	120+ years
Construction period	10–15 years

Government assessments have identified the Humber as one of the UK's significant tidal-range opportunities, with previous estimates around 1GW installed capacity and approximately 1.65TWh annual generation potential.

4. Capital Requirement

Total Project Cost

£12 Billion

Cost Breakdown

Component	Cost
Barrage structure	£5bn
Turbine systems	£2bn
Grid connection	£750m
Flood defence systems	£750m
Transport infrastructure	£1bn
Environmental mitigation	£750m
Hydrogen infrastructure	£750m
Contingency	£1bn

5. Proposed Bank of Leeds Bond Structure

Humber Tidal Infrastructure Bond

Bond Size

£2 billion

Term

30 years

Security

Backed by:

- future electricity revenues,
- infrastructure concession revenues,

- government support mechanisms,
 - project assets.
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Target Investors

UK Pension Funds

Seeking:

- inflation-linked infrastructure returns,
 - long-duration assets.
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Insurance Companies

Seeking:

- predictable cash flows.
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Institutional Investors

Seeking:

- renewable infrastructure exposure.
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6. Electricity Generation Model

Installed Capacity

Assumption:

1.2 GW

Capacity Factor

35%

Annual Generation

Approximately:

3.7 TWh/year

The Humber is assessed as having a meaningful tidal-range resource, although significantly smaller than the largest UK tidal opportunities such as the Severn.

7. Electricity Revenue Model

Base Case

Electricity price:

£100/MWh

Annual generation:

3.7 million MWh

Annual revenue:

£370 million

Higher Price Scenario

Electricity price:

£130/MWh

Revenue:

£480 million/year

8. Operating Economics

Annual operating costs:

Estimated:

£100 million

Includes:

- turbine maintenance,
- marine engineering,

- environmental monitoring,
 - operations staff.
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EBITDA

Base case:

Revenue:

£370m

Less costs:

£100m

Annual EBITDA:

£270 million

9. Bond Repayment Model

30-Year Bond Requirement

Assume:

£2bn bond issuance

Interest:

4.5%

Annual debt service:

Approximately:

£120m–£130m

Covered by:

Electricity EBITDA:

£270m

Debt coverage ratio:

Approximately:

2x

10. Additional Revenue Streams

The project should not be financed as a power station alone.

It is a regional infrastructure platform.

A. Humber Hydrogen Hub

Investment:

£750 million

Using tidal electricity:

Produce:

- green hydrogen,
- industrial fuels.

Customers:

- steel,
- chemicals,
- shipping,
- heavy transport.

Potential annual revenue:

£100m+

B. Industrial Energy Zone

Create:

Humber Renewable Industrial Cluster

Benefits:

- cheaper electricity,
- investment attraction,
- manufacturing growth.

Potential new industries:

- hydrogen steel,
 - battery materials,
 - data centres,
 - advanced manufacturing.
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C. Transport Link

Potential inclusion:

- road crossing,
- rail crossing.

Infrastructure value:

£1–2bn

D. Waterfront Regeneration

Potential:

- new housing,
 - commercial districts,
 - tourism.
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11. Yorkshire Economic Impact

Construction Phase

15-year programme.

Employment:

50,000 job-years

Permanent Employment

Direct:

5,000 jobs

Indirect:

15,000 jobs

GDP Impact

Estimated regional uplift:

£1–2 billion annually

Through:

- energy production,
 - engineering,
 - manufacturing,
 - construction,
 - logistics.
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12. Strategic Value for Yorkshire

Energy Independence

The project reduces exposure to:

- imported gas,
 - electricity price volatility.
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Industrial Renaissance

Supports:

- Sheffield engineering,
 - Hull manufacturing,
 - Humber chemicals,
 - Yorkshire logistics.
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Export Opportunity

Yorkshire could become a global leader in:

- tidal engineering,
- marine construction,

- renewable infrastructure.
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13. Risk Assessment

Construction Risk

Challenge:

Large marine infrastructure.

Mitigation:

- phased construction,
 - international engineering partnerships,
 - government guarantees.
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Environmental Risk

Potential impacts:

- sediment flows,
- fisheries,
- bird habitats.

Mitigation:

- habitat replacement,
 - fish passages,
 - environmental monitoring.
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Revenue Risk

Mitigation:

- long-term power purchase agreements,
 - regulated infrastructure model.
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14. Recommended Financing Roadmap

Phase 1 — Feasibility Bond

£250 million

Purpose:

- engineering surveys,
 - environmental studies,
 - financial modelling.
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Phase 2 — Construction Bond

£2 billion Bank of Leeds Issue

Purpose:

- initial construction,
 - turbine manufacture,
 - grid infrastructure.
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Phase 3 — Institutional Expansion

Additional:

£8–10 billion

through:

- pension funds,
 - government infrastructure finance,
 - international investors.
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Investment Recommendation

The Bank of Leeds should establish:

Humber Tidal Infrastructure Corporation

and explore issuing:

£2 Billion

30-Year Humber Renewable Infrastructure Bond

Final Investment Thesis

The Humber has historically powered Yorkshire through:

- coal,
- ports,
- chemicals,
- manufacturing.

The next century could see it powered by:

- tides,
- hydrogen,
- renewable electricity,
- advanced engineering.

A Bank of Leeds-backed Humber Tidal Infrastructure Bond would create a generational infrastructure asset:

A 120-year renewable energy project financed through a 30-year Yorkshire infrastructure bond — transforming the Humber from an industrial estuary into Britain's leading tidal energy gateway.